

MENTOR®

H - *Herbicide*

GROUP **9** HERBICIDE

A foliar applied translocated herbicide for the control of annual and perennial grass and broad-leaved weeds before sowing or planting all crops.

For use pre-harvest in cereals, oil seed rape, mustard, combining peas, field beans, linseed. For destruction of grassland, and use in stubbles, orchards, aquatic, amenity and non-crop areas. For the control of emerged weeds in industrial and amenity situations and in forestry.

Degraded by micro-organisms/microbes in the soil.

This product is a soluble concentrate containing 360 g/L glyphosate, present as 441 g/L (35.3% w/w) of the potassium salt of glyphosate.

MAPP Number 21214

MENTOR

UFI: KPF1-E0Y7-K009-RH2Q

Contains 360 g/L glyphosate, present as 441 g/L (35.3% w/w) of the potassium salt of glyphosate.

Warning

Causes serious eye irritation.

Toxic to aquatic life with long lasting effects.

Wash hands thoroughly after handling.

Wear protective gloves/protective clothing/eye protection/face protection.

Keep only in original container.

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

If eye irritation persists: Get medical advice/attention.

Dispose of contents/container to a licensed hazardous-waste disposal contractor or collection site except for empty clean containers which can be disposed of as non-hazardous waste.

To avoid risks to human health and the environment, comply with the instructions for use.



The (COSHH) Control of Substances Hazardous to Health Regulations may apply to the use of this product at work

Contents:

e5L litres

PROTECT FROM FROST

30531586a rA4 UK

DIRECTIONS FOR USE

IMPORTANT: This information is approved as part of the Product Label. All instructions within this section must be read carefully in order to obtain safe and successful use of this product.

Warnings

EXTREME CARE SHOULD BE TAKEN TO AVOID SPRAY DRIFT AS THIS CAN SEVERELY DAMAGE NEIGHBOURING CROPS OR PLANTS.

DO NOT MIX, STORE OR APPLY MENTOR IN GALVANISED OR UNLINED STEEL CONTAINERS OR SPRAY TANKS.

DO NOT leave spray mixtures in tank for long periods and make sure tanks are WELL VENTED.

Restrictions

A period of at least 6 hours and preferably 24 hours rain-free must follow application of MENTOR.

Do not spray onto weeds which are naturally senescent, or where growth is impaired by drought, high temperatures, a covering of dust, flooding or frost at, or immediately after application, otherwise poor control may result.

Do not spray in windy conditions as drift onto desired crops or vegetation could severely damage or destroy them.

After application, large concentrations of decaying foliage, stolons, roots or rhizomes should be dispersed or buried by thorough cultivation before crop drilling.

Applications of lime, fertilizer, farmyard manure and pesticides should be delayed until 5 days after application of MENTOR. TREATED POISONOUS PLANT SPECIES MUST BE REMOVED BEFORE REGRAZING OR CONSERVING.

Where Ragwort is present users should consult the Code of Practice on How to Prevent the Spread of Ragwort. Ragwort plants sprayed with this herbicide are more palatable and contain higher levels of toxins.

Animals should be excluded from treated areas until any Ragwort has completely recovered or died and there is no visible sign of the dead weed. Do not include treated Ragwort in hay or silage crops..

Weeds controlled

MENTOR is a foliar acting herbicide which controls annual and perennial grasses and most broadleaved weeds when used as directed. It is important that all weeds are at the correct growth stage when treated, otherwise some re-growth may occur and this will need re-treatment.

Apply MENTOR herbicide once grasses and broadleaved weeds have emerged and they have ACTIVELY GROWING green leaves

- PERENNIAL GRASSES must have a full emergence of healthy, green leaf. (Common Couch, for example, becomes susceptible at the onset of tillering and new rhizome growth commences which usually occurs when plants have 4 – 5 leaves, each with 10 – 15 cm of new growth).
- PERENNIAL BROAD-LEAVED WEEDS are most susceptible around the flowering stage.
- ANNUAL GRASSES AND BROAD-LEAVED WEEDS should have at least 5 cm of leaf, or 2 expanded true leaves, respectively. In set-aside, annual grasses are best treated at full ear emergence, or before stem elongation. Application during stem extension phase of annual grasses e.g. Black-grass and Brome species on set-aside between the end of April and end of May, may result in poor control and require re-treatment.

- OTHER SPECIES - recommendations for specific Areas of Use are given in the Recommendation Tables, pages 2 – 11.
- This product will not give an acceptable level of control of Horsetails (*Equisetum arvense*) – repeat treatment will be necessary.

Following crops

Upon soil adsorption the herbicidal properties of MENTOR are lost permitting the drilling of crops 48 hours after application. Planting of trees, shrubs etc may take place 7 days after application. Grass seed may be sown from 5 days after treatment; see the 'Recommendation Tables' for specific restrictions on direct drilled crops.

Weed resistance strategy

There is low risk for the development of weed resistance to MENTOR. Strains of some annual weeds (e.g. Black-grass, Wild oats and Italian Ryegrass) have developed resistance to herbicides which may lead to poor control. A strategy for preventing and managing such resistance should be adopted. This should include integrating herbicides with a programme of cultural control measures. Guidelines have been produced by the Weed Resistance Action Group and copies are available from the HGCA, CPA, your distributor, crop adviser or product manufacturer (Bayer).

#Crop Specific Information

COMPLIANCE WITH THE FOLLOWING CONDITIONS OF USE AND ALL SAFETY PRECAUTIONS MARKED* IS A LEGAL REQUIREMENT

Crops/situations	Maximum individual dose (litres product/ hectare):	Maximum total dose (litres product/ hectare/ crop/ annum):	Latest time of application:
Pre-harvest Winter wheat, winter barley, winter oats, spring wheat, spring barley, spring oats, durum wheat, combining pea, field bean	4.0	4.0	7 days before harvest
Post planting and pre-emergence of listed cereals, oilseed rape, combining peas, vining peas, field beans, mustard, linseed, sugar beet, swede, turnip, bulb onion and leek	1.5	1.5	Pre-emergence
Oilseed rape and linseed	4.0	4.0	14 days before harvest
Mustard	4.0	4.0	8 days before harvest
All edible crops (stubble), all non-edible crops (stubble)	5.0 or 1.5	5.0 or 4.0	5 days before drilling or planting of the following crop 2 days before the drilling or planting of the following crop or 24 hours before cultivating
All edible and non-edible crops (destruction, before sowing/planting).	5.0	-	-
Grassland	6.0	6.0	5 days before harvest, grazing or drilling
Apple and pear orchards	5.0	5.0	After harvest but before green cluster stage
Cherry, plum and damson orchards.	5.0	5.0	After harvest (post leaf fall) but before white bud stage
Green cover on land not being used for crop production	6.0	6.0	24 hours before cultivating
Non-crop including natural surfaces not intended to bear vegetation, permeable surfaces overlying soil, hard surfaces	5.0	-	-
Amenity vegetation	5.0	-	-

- Tractor Mounted: To avoid foaming do not use top tank agitation. Half fill the spray tank with clean water, start gentle agitation, then add the correct amount of MENTOR. Top up the tank with water to the required level. Use of a defoamer may be necessary.

b) Rotary Atomisers – for use in orchards

When rotary atomisers are used to apply MENTOR ensure that the droplet diameter falls within the range 200-300 microns for all uses.

c) Hand-held Wipers

MENTOR may be applied through the weed wiper mini. Use a concentration of 1 part MENTOR to 2 parts of water and add a scarlet dye if required. Care should be taken to avoid dripping onto wanted vegetation.

d) Cut Stump Application

Enso attachment to rotary saws: This technique is specific to scrub clearance in forestry. A water-soluble dye may be added to MENTOR to help identify treated stumps.

- e.g. Knapsack Hypro 1.2 – 2.4
Cooper Pegler Floodjet green, red
- Tractor Hypro 04-F110, 08-F110
Tee Jet 11004, 11008

Low Volume Application (minimum 80 L/ha) Low volume application can be achieved by reducing pressure and the appropriate nozzle selection. Low drift nozzles which produce a medium spray quality (BCPC definition) should be used to minimise the risk of drift:

- e.g. Knapsack Cooper Pegler VLV 100
Hypro AN 1.0
- Tractor Hypro Lo Drift

Filling the Sprayer

- Knapsack: Half fill the spray tank with clean water, add the correct amount of MENTOR and top up with water. Mix thoroughly.

HAND-HELD EQUIPMENT: SPECIFIC GUIDANCE

e) Knapsack Sprayer Applicators

When used at a walking speed of 1 m/sec to apply a swath of 1 m width, most knapsack sprayers deliver 200 L/ha spray volume (or 10 litres per 500 m²). To apply 4.0 L/ha of MENTOR, therefore, use a 2% solution (e.g. 200 ml MENTOR made up to 10 litres).

When used as above, knapsack sprayers fitted with low volume nozzles typically deliver 100 L/ha spray volume (or 10 litres per 1000 m²). To apply 4.0 L/ha MENTOR in this case, use a 4% solution.

Filling the sprayer – hand-held machines Stir the correct amount of MENTOR into the sprayer half filled with clean water. Top up with water, close the top and shake gently to ensure good mixing.

f) Spot Gun Applicators – for treatment of individual weeds

Apply 5 ml of spray to target weed using narrow cone TG-3 or TG-5 nozzle.

Spot Diameter (metres)	Amount of MENTOR (ml) per 5 litres of spray solution for targeted dosages of		
	3.0 L/ha	4.0 L/ha	5.0 L/ha
0.3	20	28	35
0.6	85	110	140

QUICK GUIDE FOR HAND-HELD SPRAYERS

Knapsack sprayers should be calibrated by individual users, but as a guide where standard nozzles* are fitted the dilutions in the table below should apply.

	Rate in litres per hectare	Dilution in knapsack sprayer (mls per litre of water)	Dilution factor	Area covered by 10 litres of spray solution
Annuals & seedlings only	3	15	1 in 67	500 m²
Perennials present	5	25	1 in 40	500 m²

* Applying 200 L/ha when walking at 1 metre per second

Compatibility

Do not tank mix MENTOR with adjuvants, pesticides or fertilisers except as advised by Bayer Crop Science. For up to date information on compatible products contact Bayer CropScience Limited.

MENTOR is compatible with Mixture B (ADJ 0161). Where conventional hydraulic sprayers are being used Mixture B may be added to the spray tank solution, at a rate of 2% of the final water volume, for all pre-plant and post-plant directed sprays only.

DO NOT APPLY WITH MIXTURE B TO EDIBLE CROPS, OR GRASSLAND WEEDS.

Do not tank-mix MENTOR when using rotary atomiser sprayers.

For hydraulic sprayers: maintain continuous agitation when using MENTOR in tank mixture.

For knapsack sprayers: mix thoroughly and use immediately when using MENTOR in tank mixture.

COMPANY ADVISORY INFORMATION

This section is not part of the Product Label under the Plant Protection Products Regulations 1995 and provides additional advice on the product.

General Information

MENTOR is an advanced glyphosate formulation. To maximise the safe use of MENTOR to operator, consumer and environment, the label recommendations and the DEFRA/HSC publication “Code of Practice for using Plant Protection Products, 2006” should be adhered to.

MENTOR herbicide is a foliar-acting herbicide with broad-spectrum activity. It is taken up by foliage and translocated to underground roots, rhizomes and stolons, providing control of both annual and perennial grasses and broad-leaved weeds. MENTOR is rapidly adsorbed onto particulate

Sprayer Hygiene

It is essential to thoroughly clean-out spray tanks, pumps and pipelines and nozzle or disc assemblies, with a recommended detergent cleaner, between applying this product and other pesticides to avoid contamination from pesticide residues.

Disposal

Follow the guidance on the disposal of surplus spray solution, tank washings, concentrate and containers as given in Section 5 of DEFRA/HSC publication “Code of Practice for using Plant Protection Products, 2006

Sprayer Hygiene

The product must only be transferred and measured using a closed transfer system that meets or exceeds British Standard BS 6356 Part 9. If any doubt exists regarding equipment suitability contact the equipment manufacturer or your agronomist for further advice.

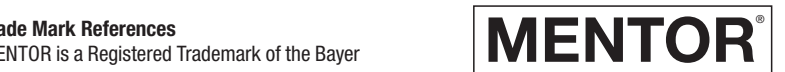
Remove the tamper evident plastic cap from valve unit fitted to the top of the IBC. Attach the coupler of the transfer system as instructed in the equipment manual to the IBC and operate using the instructions ensuring that the transfer system is clean and empty before placing in the storage position.

The container must be empty of product and left in a clean condition for collection. Do not attempt to rinse the inside - external clean only.

Trade Mark References

MENTOR is a Registered Trademark of the Bayer Group.

All other brand names referred to are trademarks of other manufacturers in which proprietary rights may exist.



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COMPLIANCE WITH THE FOLLOWING CONDITIONS OF USE AND ALL SAFETY PRECAUTIONS MARKED * IS A LEGAL REQUIREMENT FOR PROFESSIONAL USE ONLY AN AGRICULTURAL/HORTICULTURAL/FORESTRY HERBICIDE

Crops/situations:
Wheat, (including Durum wheat), barley, oats, combining pea, vining pea, field bean; Oilseed rape, mustard, linseed; Sugar beet, swede, turnip, bulb onion, leek; All edible crops (stubble), all non-edible crops (stubble); All edible and non-edible crops (destruction, before sowing/planting); Grassland; Apple, pear, plum, cherry damson; Green cover on land not being used for crop production; Farm non-crop areas including natural surfaces not intended to bear vegetation, permeable surfaces overlying soil, hard surfaces; Amenity vegetation; Forest nursery, forest (weed control, stump application and chemical thinning).

Maximum individual dose:	}	Full details are given in the Statutory Area on the attached leaflet (see Crop Specific Information – marked #)
Maximum number of treatments:	}	
Latest time of application:	}	
Other specific restrictions:	}	

READ ALL OTHER SAFETY PRECAUTIONS AND DIRECTIONS FOR USE BEFORE USE

SAFETY PRECAUTIONS

- Operator protection Engineering control of operator exposure must be used where reasonably practicable in addition to the following personal protective equipment:
 - *WEAR SUITABLE PROTECTIVE GLOVES AND FACE PROTECTION (FACESHIELD) when handling the concentrate and when handling contaminated surfaces.
 - *WEAR SUITABLE PROTECTIVE CLOTHING (COVERALLS), SUITABLE PROTECTIVE GLOVES AND RUBBER BOOTS when using hand-held sprayers and hand-held rotary atomisers.
 - *WEAR SUITABLE PROTECTIVE CLOTHING (COVERALLS), SUITABLE PROTECTIVE GLOVES, RUBBER BOOTS AND FACE PROTECTION (FACESHIELD) when using weedwiper equipment and cut stump treatments.
 - * However, engineering controls may replace personal protective equipment if a COSHH assessment shows they provide an equal or higher standard of protection.
- Environmental protection Do not contaminate water with the product or its container except when used as directed. Do not clean application equipment near surface water. Avoid contamination via drains from farmyards and roads.
- Storage and disposal KEEP AWAY FROM FOOD, DRINK AND ANIMAL FEEDINGSTUFFS. KEEP OUT OF REACH OF CHILDREN. KEEP IN ORIGINAL CONTAINER, tightly closed, in a safe place. EMPTY CONTAINER COMPLETELY and dispose of safely. RINSE CONTAINER THOROUGHLY by using an integrated pressure-rinsing device or manually rinse three times. Add washings to sprayer at time of filling and dispose of safely. Triple rinsed containers may be disposed of as non-hazardous waste.

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RECOMMENDATION TABLES

AREA OF USE	TARGET WEEDS/USAGE	CROP/SITUATION	WEED INFESTATION	APPLICATION RATE L/ha	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
PRE-HARVEST ARABLE CROPS	Common Couch	WINTER and SPRING WHEAT, DURUM WHEAT, WINTER and SPRING BARLEY and WINTER and SPRING OATS	1 to 25 shoots/m ² Up to 75 shoots/m ² Over 75 shoots/m ²	2.0 3.0 4.0	80-250 L/ha*	Grain/seed moisture must not exceed 30% at spraying. Harvest intervals: CEREALS, PEAS, BEANS 7+ days OILSEED RAPE 14-21 days LINSEED 14-28 days MUSTARDS 8-10 days Use high clearance, narrow wheeled tractors, wide booms and crop dividers. DO NOT TREAT CROPS GROWN FOR SEED. Effects on brewing and baking have not been tested. Consult grain merchant or processor before use. Treated straw must not be used as a horticultural mulch. *Rotary atomisers may be used at a water volume of 40 L/ha. Ensure droplet diameter falls within the range 200-300 microns. # Use higher volumes for dense canopies.
		OILSEED RAPE MUSTARDS	Up to 75 shoots/m ² Over 75 shoots/m ²	3.0 4.0	100-250 L/ha#	
		COMBINING PEAS FIELD BEANS	Up to 75 shoots/m ² Over 75 shoots/m ²	3.0 4.0	80-250 L/ha*	
		LINSEED	Up to 75 shoots/m ² Over 75 shoots/m ²	3.0 4.0	80-250 L/ha*	
		Perennial broad-leaved weeds and other perennial grasses	WINTER and SPRING WHEAT, DURUM WHEAT, WINTER and SPRING BARLEY and WINTER and SPRING OATS	All levels/species	80-250 L/ha*	
			OILSEED RAPE MUSTARDS	All levels/species	200-250 L/ha	
			COMBINING PEAS FIELD BEANS	All levels/species	80-250 L/ha*	
	Annual weeds, prior to direct combining	OILSEED RAPE MUSTARDS	All levels/species	3.0	100-250 L/ha#	
			LINSEED	All levels/species	80-250 L/ha	
ALL EDIBLE AND NON-EDIBLE CROPS (DESTRUCTION, BEFORE SOWING/ PLANTING)	Vegetation management	-	Annual weeds Perennial grasses Perennial broad-leaved weeds	1.5 4.0 5.0	80-250 L/ha* or hand-held equipment (See mixing and spraying section)	Do not use under polythene or glass. *Rotary atomisers may be used at a water volume of 40 L/ha. Ensure droplet diameter falls within the range 200-300 microns Apply the annual weed dose at least 2 days before sowing/planting. Apply at perennial weed doses at least 5 days before sowing/planting. Do not use in or alongside hedgerows
POST SOWING/ PLANTING, PRE-EMERGENCE OF THE CROP	Volunteer cereals and annual weeds	LISTED CEREALS OILSEED RAPE, MUSTARD, LINSEED, PEAS, FIELD BEANS, SUGAR BEET, SWEDE, TURNIP, ONION and LEEK	All levels/species	1.5	80-250 L/ha*	CAUTION – Ensure that spraying precedes ANY crop emergence. * Rotary atomisers may be used at a water volume of 40 L/ha. Ensure droplet diameter falls within the range 200-300 microns.

TARGET WEEDS/USAGE	CROP/SITUATION	WEED INFESTATION	APPLICATION RATE L/HA	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
Common Couch	BEFORE ALL CROPS EXCEPT ORCHARDS	Up to 75 shoots/m ² Over 75 shoots/m ²	3.0 4.0	80-250 L/ha*	Do not cultivate immediately before spraying. For PERENNIAL weed control, allow: – 21+ days growth before spraying in spring – VOLUNTEER POTATOES to make ample top growth – 5 days before cultivating or drilling For ANNUAL weed control, allow: – 24 hours before cultivating – 48 hours before drilling Allow 7 days before planting trees * Rotary atomisers may be used at a water volume of 40 L/ha. Ensure droplet diameter falls within the range 200-300 microns.
Other perennial grasses; volunteer potatoes (autumn only)		All levels of all species	4.0		
Perennial broad-leaved weeds		All levels of all species	5.0		
Volunteer cereals and annual weeds		All levels of all species	1.5		
Perennial grasses and broad-leaved weeds		Arable weeds Pasture weeds	4.0 5.0		
Common Couch	BEFORE or DURING REMOVAL FROM PRODUCTION	Up to 75 shoots/m ² Over 75 shoots/m ²	3.0 4.0	80-250 L/ha* or hand-held equipment or tractor mounted weed wiper	Before using on land taken out of production as part of a grant aided scheme, ensure compliance with the management rules of that scheme. Do not 'top' or cultivate immediately before application. For PERENNIAL weed control, allow:- – 21+ days growth before spraying in spring – 5 days before cultivating or drilling. For ANNUAL weed control, allow: – 24 hours before cultivating. Do not direct drill after set aside. * Rotary atomisers may be used at a water volume of 40 L/ha. Ensure droplet diameter falls within the range 200-300 microns. Avoid applications during stem elongation as reduced control and re-spray is likely. Best control of annual grasses is achieved between full ear emergence and senescence. +Only for weeds listed as per grassland destruction application rate table below
Perennial broad-leaved weeds and other perennial grasses		All levels/species	4.0		
Annual weeds: Early autumn/spring Late spring/summer		All levels/species All levels/species	1.5 3.0		
		Annual weeds only Perennial grasses Perennial broad-leaved weeds Perennial broad-leaved weeds as listed below.	3.0 4.0 5.0 6.0+		
Natural regeneration and cover crop destruction		AFTER SHORT ROTATION or LONG TERM REMOVAL FROM PRODUCTION			
Short rotation Ryegrass, longer leys and permanent pasture	GRASS	Short rotation Ryegrass with annual weeds Leys 2-4 years old with perennial grass weeds Long leys 4-7 years old with perennial broad-leaved weeds Permanent pasture See Weed Table	3.0 4.0 5.0 6.0	150-250 L/ha	Treat EITHER before grazing/mowing in June-Oct, when growth is 30-60 cm, not dense and lacking mature seeds, OR re-growth after grazing/mowing. Select the application rate which controls/destroys the least susceptible weed and grass species present in the sward. Grass may be conserved or grazed by cattle, dairy cows or sheep 5+ days after spraying. REMOVE POISONOUS PLANTS BEFORE GRAZING/MOWING. If Ragwort is present, the guidance in the 'DIRECTIONS FOR USE' must be followed. ONLY direct drill grass and clover EITHER into 1-2 year leys without mat, 5+ days after spraying, OR long leys with some mat, in the spring following autumn application.

APPLICATION RATE FOR GRASSLAND DESTRUCTION							
3.0 L/ha		4.0 L/ha		5.0 L/ha		6.0 L/ha	
Common Chickweed	Black-bent	Broad-leaved Dock	Bracken** Common Nettle	Creeping Buttercup	Common Ragwort	Hard Rush	
Dock Seedlings	Cock's-foot	Common Bent		Daisy	Heath Rush	Jointed Rush	
Mayweed species	Common Couch	Creeping Bent		Perennial Sow-thistle	<i>Molinia</i> (Purple Moor-grass)	<i>Nardus</i> (Mat grass)	
Meadow Foxtail	Creeping Soft-grass	Curled Dock		Sedges	Red Fescue	Sheep's Fescue	
Speedwell species	Perennial Rye-grass	Plantains		Soft Rush	White Clover*	Yellow Rattle	
	Soft Brome	Yorkshire Fog		Tufted Hairgrass			
			Common Sorrel				
			Creeping Thistle				
			Dwarf Thistle				
			Red Clover				
			Sheep's Sorrel				
			Spear Thistle				
			Yarrow				

* White Clover is best cut in June and sprayed one month later.

** At full frond expansion

AREA OF USE	TARGET WEEDS/USAGE	CROP/SITUATION	WEED INFESTATION	APPLICATION RATE L/HA	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
ORCHARDS	Perennial grasses and broad-leaved weeds	APPLE, PEAR, PLUM, CHERRY, DAMSON	All levels of most species	5.0	Hydraulic sprayers including hand held 200-400 L/ha or Rotary atomizers at 40 L/ha	Spray AFTER autumn leaf-fall and BEFORE: Apples, pears – green cluster stage Stone fruit – white bud stage Treat root suckers in late spring only. Trees must have been established 2+ years. Avoid contact with tree 30+ cm above ground.
	-		5.0			
IN-CROP (TRACTOR-MOUNTED WEED WIPER) APPLICATION)	Bolters, weed beet, other weeds	ARABLE CROPS AND GRASSLAND SET ASIDE	All levels	1:1 dilution with water OR 1:2 dilution with water in hot, dry conditions. For 'new generation' wipers consult the manufacturer for guidance.		Weeds must be 10+ cm taller, and wiper 5+ cm higher, than desired vegetation. Wipe dense populations twice, in opposite directions. BOLLWORM applications, 2 weeks apart, from early July to early August. Contact Bayer or your distributor for specific recommended weed wiper applicators. <i>POISONOUS WEEDS</i> and grazing/mowing interval – See GRASSLAND section. If Ragwort is present, the guidance in the 'DIRECTIONS FOR USE' must be followed.
NATURAL SURFACES NOT INTENDED TO BEAR VEGETATION, PERMEABLE SURFACES OVERLYING SOIL. ALL SITUATIONS (DESTRUCTION, BEFORE PLANTING).	Vegetation management	Including farmyards roadsides, paths, and along fences and walls	Annual weeds Perennial grasses and broad-leaved weeds	1.5 4.0–5.0	Hydraulic sprayers (boom and knapsack) at water volumes 80-400 L/ha or rotary atomisers* at water volumes 40 L/ha or hand-held equipment. See Mixing & Spraying section.	Do not use under polythene or glass. * Where rotary atomisers are used their droplet diameter must fall within the range 200-300µm
HARD SURFACES	Vegetation management	Including farmyards roadsides, paths, hard surfaces and along fences and walls	Annual weeds Perennial grasses and broad-leaved weeds	1.5 4.0–5.0	Hydraulic sprayers (boom and knapsack) at water volumes 80-400 L/ha or rotary atomisers* at water volumes 40 L/ha or hand-held equipment. See Mixing & Spraying section.	Apply this product carefully. Ensure spraying takes place only when weeds are actively growing (normally March to October) and is confined to only visible weeds including those in the 30 cm swath covering the kerb edge and road gully – do not overspray drains.
AMENITY VEGETATION	Vegetation management	Areas of semi-natural or ornamental vegetation including trees. Areas of bare soil around ornamental plants or areas intended for ornamental planting or clearance of allotments	Annual weeds Perennial grasses and broad-leaved weeds	1.5 4.0-5.0	80-400 L/ha*	Hydraulic sprayers, rotary atomisers or weed wipers may be used. DO NOT USE IN OR ALONGSIDE HEDGEROWS. DO NOT USE UNDER POLYTHENE OR GLASS.
	Root suckers	-	All species	5.0		

FORESTRY/FARM FORESTRY WEED CONTROL

MENTOR can be used for site preparation and for weed control in planted out trees

AREA OF USE	TARGET WEEDS/USAGE	WEED INFESTATION	APPLICATION RATE L/ha	WATER VOLUME	APPLICATION TIMING AND GUIDANCE
FORESTRY: - PRE-PLANTING	Arable land, planting, replanting, & grassland areas	Arable weeds Grassland weeds	4.0 5.0	Hydraulic sprayers: 80-250 L/ha or rotary atomisers: 40 L/ha*	All tree species may be planted 7 days or more after treatment. *Where rotary atomisers are used their droplet diameter must fall within the range 200-300µm.
FORESTRY: - POST-PLANTING (DIRECTED) IN CONIFERS & BROAD-LEAVED TREES	Clean-up around trees with knapsack applicators	Annual/perennial grasses and broad-leaved weeds	4.0	Hand held equipment. Knapsack: Apply as a 2% concentration or Weed wiper mini: apply as a concentration of 1 part MENTOR to 2 parts water (see Mixing & Spraying section)	It is ESSENTIAL to use a TREE GUARD for all applications made in the growing season. Treat bracken after frond tips are unfurled but before senescence. Treat heather late August to end September. All other woody weeds are treated June to August, before leaf senescence but after new growth of crop has hardened. (*) For improved control of Rhododendron apply 6.4 L/ha MENTOR, adding Mixture B (ADJ A0161) at 2% of spray volume. Application using a weed wiper is not suitable.
		Woody weeds: Bracken/Beech Brush/Brambles Sycamore/Oak Hazel/Willow/Ash (excluding Rhododendron)	3.0		
		Heather (peat soils) Heather (mineral soils) Rhododendron (*)	4.0 6.0 10.0		
FORESTRY: - POST-PLANTING (OVERALL DORMANT SEASON IN CERTAIN CONIFERS – CONIFER RELEASE)	Grass weeds: - Lowland areas - Upland areas Bracken Beech & Birch Brambles	Black Bent, Cock'sfoot, Common Couch, Creeping Soft-grass, False Oat-grass, Fescues, Meadow-grasses, other Bent species, Purple Moor-grass, Sweet Vernal-grass, Tufted Hair-grass, Wavy Hair-grass, Wood Small-reed (Bush grass) All levels of all species All levels of all species All levels of all species	1.5 2.0 2.0 2.0 3.0	Hydraulic sprayers: 200-250 L/ha or hand-held equipment – see 'Mixing and Spraying' section	DO NOT OVERALL SPRAY trees being grown for ORNAMENTAL PURPOSES, including CHRISTMAS TREES. Species safe to spray when fully dormant and leader growth has hardened: Corsican, Lodgepole and Scots Pine, Norway Spruce, Sitka Spruce, Lawson Cypress, Western Red Cedar. Douglas Fir and Noble Fir – safe to spray when fully dormant and leader growth has hardened but NOT in spring. If overall application takes place after the optimum timing weed control may be reduced. It is advisable to spray a limited area of forest to test crop safety under local conditions before widespread overall application in subsequent years. These recommended application rates refer to forestry usage only. Inadequate control may result if used in other areas. Caution: The timing of hardening of leader growth varies considerably between locations and between seasons. It may occur as early as the end of July or be delayed to October or later. To avoid damage to Lammas growth, sprays should be directed away from leaders.
FORESTRY: - STUMP APPLICATION FOR CHEMICAL THINNING	Deciduous trees Coniferous trees	All species All species	10% solution of MENTOR in water 20% solution of MENTOR in water		Apply the solution to saturate the rim of the newly cut surface, with a suitably adapted clearing saw, spot gun or paintbrush. Treat as soon as possible after felling, in the period November to March/April. Do not apply in the period of active sap flow in the spring/early summer. Do not cut trenches or drill holes and fill with the solution or use undiluted product. Note: for ease of identification of treated areas a suitable, commercially available, water-soluble dye may be added to the prepared spray solution
FORESTRY: - CHEMICAL THINNING BY INJECTION OF TREE STEMS	Coniferous and deciduous species	-	2 ml neat MENTOR per cut per 10 cm diameter (or less) of tree		Use a hatchet to cut one notch in trees up to 10 cm diameter and apply 2 ml of the solution to each cut. Use two or three notches in trees over 10 cm diameter. Do not treat in the period of active sap flow in the spring/early summer.